

$$\textcircled{2} \int n \sin \frac{n}{2} \, dn.$$

Let $u = n$ and $v = \sin \frac{n}{2}$.

$$u \int v \sin \frac{n}{2} = \int \left[\frac{du}{dn} \right] dn \cdot \int \left[\frac{dv}{dn} \right] dn.$$

$$n \int \sin \frac{n}{2} = \int 1 \cdot \int \sin \frac{n}{2} \, dn.$$

$$n \cdot 2 (-\cos(\frac{n}{2})) - \int 1 \cdot -2 \cos(\frac{n}{2})$$

$$n - 2 \cos(\frac{n}{2}) + 2 \int 2 \sin(\frac{n}{2}) + c$$

$$n(-2) \cos(\frac{n}{2}) + 4 \sin \frac{n}{2} + c$$

$$\left(-2n \cos \frac{n}{2} + 4 \sin(\frac{n}{2}) + c \right)$$

Final Answer